

Assignment 1: Snowflake Environment Setup (Beginner)

Objective: Understand Snowflake UI and basic objects.

Tasks

1. Log in to Snowflake (trial account).
2. Create:
 - A **Database**
 - A **Schema**
 - A **Virtual Warehouse**
3. Show all databases and schemas.
4. Suspend and resume the warehouse.

Deliverable:

- SQL script
 - Screenshot of created objects
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Assignment 2: User, Role & Access Control (RBAC)

Objective: Learn Snowflake security and role hierarchy.

Tasks

1. Create roles:
 - STUDENT_ROLE
 - INSTRUCTOR_ROLE
2. Create users:
 - student1
 - instructor1
3. Assign:
 - STUDENT_ROLE → student1
 - INSTRUCTOR_ROLE → instructor1
4. Grant permissions:
 - Student → **SELECT** on tables
 - Instructor → **SELECT, INSERT**
5. Verify access using SHOW GRANTS.

Deliverable:

- SQL commands
 - Short explanation of RBAC
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Assignment 3: Table Creation & Data Loading

Objective: Work with structured data.

Tasks

1. Create a table:

```
CREATE TABLE students (
```

```
  id INT,
```

```
  name STRING,
```

```
  course STRING,
```

```
  score INT
```

```
);
```

2. Insert at least **10 records**.
3. Query:
 - All students
 - Students with score > 80
4. Update one record.
5. Delete one record.

Deliverable:

- SQL file with queries
 - Output screenshots
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Assignment 4: File Loading using Stage & COPY (Intermediate)

Objective: Learn Snowflake data ingestion.

Tasks

1. Create an **internal stage**.
2. Upload a CSV file.
3. Use COPY INTO to load data.
4. Validate loaded records.
5. Handle one data error using VALIDATION_MODE.

Deliverable:

- SQL script
 - CSV file
 - Query output
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Assignment 5: Time Travel & Data Recovery

Objective: Understand Snowflake Time Travel.

Tasks

1. Delete records from a table.
2. Query data using **Time Travel**.
3. Restore deleted data.
4. Explain how Time Travel helps in recovery.

Deliverable:

- SQL queries
 - Short explanation
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Assignment 6: Warehouses & Performance

Objective: Learn compute management.

Tasks

1. Create:
 - SMALL_WH
 - MEDIUM_WH
2. Run same query on both warehouses.
3. Compare execution time.
4. Suspend unused warehouse.

Deliverable:

- Query history screenshot
 - Performance comparison notes
-

Assignment 7: Views & Secure Views

Objective: Control data visibility.

Tasks

1. Create a **View** hiding sensitive columns.
2. Create a **Secure View**.
3. Grant view access to Student role.
4. Test access using Student role.

Deliverable:

- SQL queries
 - Explanation of Secure View
-

Assignment 8: Scaling & Auto Suspend (Advanced)

Objective: Understand Snowflake scaling behavior.

Tasks

1. Enable **auto-suspend** and **auto-resume**.
2. Configure **multi-cluster warehouse**.
3. Simulate concurrent queries.
4. Observe scaling behavior.

Deliverable:

- Warehouse configuration
- Query history evidence